

Version control with Git

Week 1

"FINAL".doc



FINAL.doc!



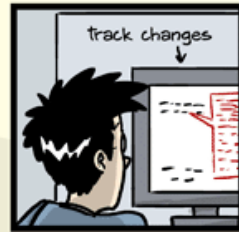
FINAL_rev.2.doc



FINAL_rev.6.COMMENTS.doc



FINAL_rev.8.comments5.
CORRECTIONS.doc



FINAL_rev.18.comments7.
corrections9.MORE.30.doc



FINAL_rev.22.comments49.
corrections.10.#@\$%WHYDID
ICOMETOGRADSCHOOL?????.doc

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www.phdcomics.com

Version control with Git

Other version control systems



Git



Text

Binary

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Version control with Git

```
$ git diff
```

```
diff --git a/CONTRIBUTING.md b/CONTRIBUTING.md
```

```
index 8ebb991..643e24f 100644
```

```
--- a/CONTRIBUTING.md
```

```
+++ b/CONTRIBUTING.md
```

```
@@ -65,7 +65,8 @@ branch directly, things can get messy.
```

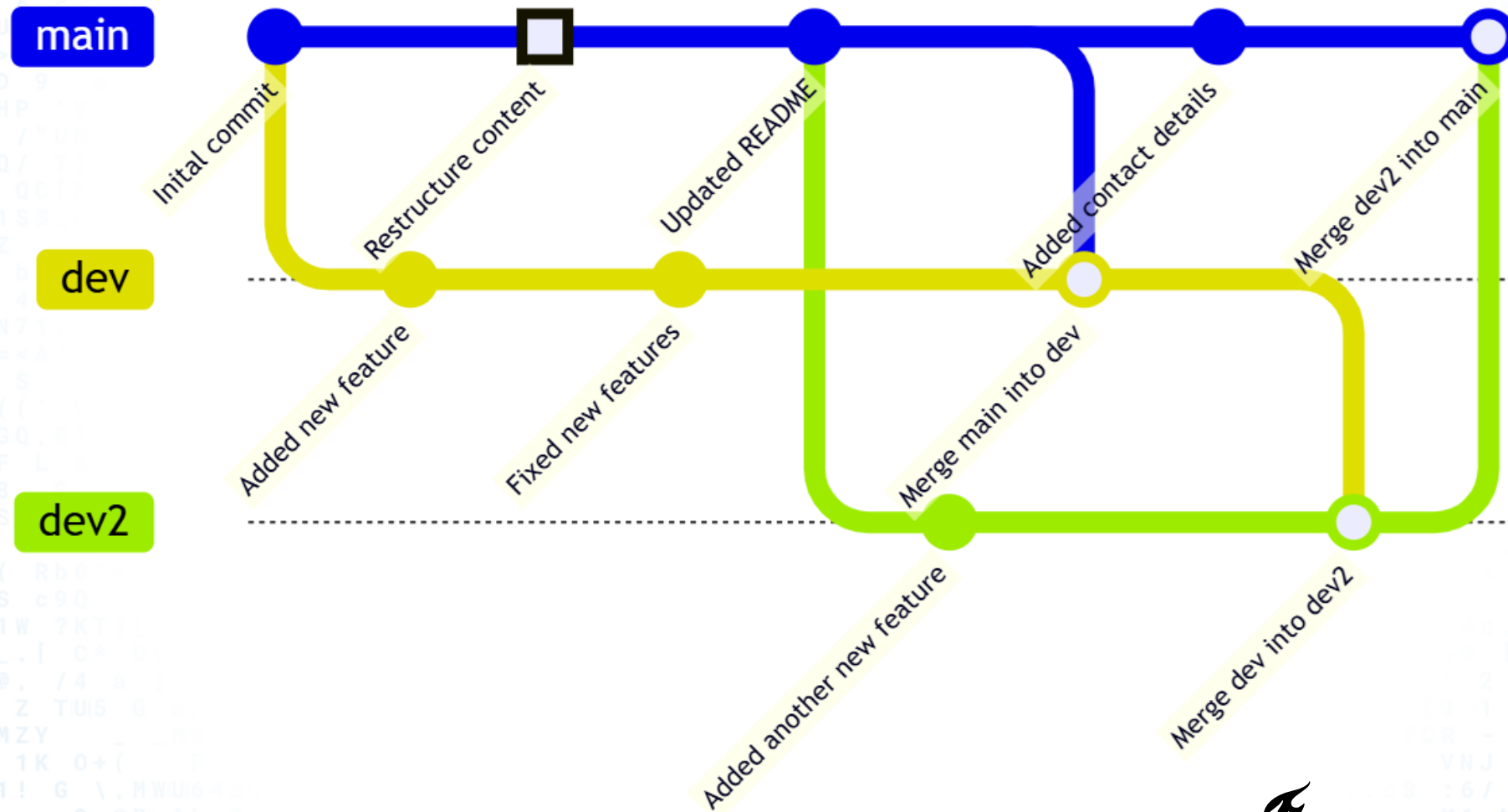
Please include a nice description of your changes when you submit your PR;
if we have to read the whole diff to figure out why you're contributing
in the first place, you're less likely to get feedback and have your change

~~merged in.~~

+merged in. Also, split your changes into comprehensive chunks if your patch is
+longer than a dozen lines.

If you are starting to work on a particular area, feel free to submit a PR
that highlights your work in progress (and note in the PR title that it's

Version control with Git



THIS IS GIT. IT TRACKS COLLABORATIVE WORK
ON PROJECTS THROUGH A BEAUTIFUL
DISTRIBUTED GRAPH THEORY TREE MODEL.

COOL. HOW DO WE USE IT?

NO IDEA. JUST MEMORIZE THESE SHELL
COMMANDS AND TYPE THEM TO SYNC UP.
IF YOU GET ERRORS, SAVE YOUR WORK
ELSEWHERE, DELETE THE PROJECT,
AND DOWNLOAD A FRESH COPY.



Version control with Git



Git Cheat Sheet

Getting Started

Start a new repo:
`git init`

Clone an existing repo:
`git clone <url>`

Prepare to Commit

Add untracked file or unstaged changes:
`git add <file>`

Add all untracked files and unstaged changes:
`git add .`

Choose which parts of a file to stage:
`git add -p`

Move file:
`git mv <old> <new>`

Delete file:
`git rm <file>`

Tell Git to forget about a file without deleting it:
`git rm --cached <file>`

Unstage one file:
`git reset <file>`

Unstage everything:
`git reset`

Check what you added:
`git status`

Make Commits

Make a commit (and open text editor to write message):
`git commit`

Make a commit:
`git commit -m 'message'`

Commit all unstaged changes:
`git commit -am 'message'`

Move Between Branches

Switch branches:
`git switch <name>`

OR
`git checkout <name>`

Create a branch:
`git switch -c <name>`

OR
`git checkout -b <name>`

List branches:
`git branch`

List branches by most recently committed to:
`git branch --sort=-committerdate`

Delete a branch:
`git branch -d <name>`

Force delete a branch:
`git branch -D <name>`

Diff Staged/Unstaged Changes

Diff all staged and unstaged changes:
`git diff HEAD`

Diff just staged changes:
`git diff --staged`

Diff just unstaged changes:
`git diff`

Diff Commits

Show diff between a commit and its parent:
`git show <commit>`

Diff two commits:
`git diff <commit> <commit>`

Diff one file since a commit:
`git diff <commit> <file>`

Show a summary of a diff:
`git diff <commit> --stat`
`git show <commit> --stat`

Ways to refer to a commit

Every time we say <commit>, you can use any of these:

* a branch	main
* a tag	v0.1
* a commit ID	3e887ab
* a remote branch	origin/main
* current commit	HEAD
* 3 commits ago	HEAD^^^ or HEAD~3

Discard Your Changes

Delete unstaged changes to one file:
`git restore <file>`

OR
`git checkout <file>`

Delete all staged and unstaged changes to one file:
`git restore --staged --worktree <file>`

OR
`git checkout HEAD <file>`

Delete all staged and unstaged changes:
`git reset --hard`

Delete untracked files:
`git clean`

'Stash' all staged and unstaged changes:
`git stash`

Edit History

"Undo" the most recent commit (keep your working directory the same):
`git reset HEAD^`

Squash the last 5 commits into one:
`git rebase -i HEAD~5`

Then change "pick" to "fixup" for any commit you want to combine with the previous one

Undo a failed rebase:
`git reflog BRANCHNAME`

Then manually find the right commit ID in the reflog, then run:
`git reset --hard <commit>`

Change a commit message (or add a file you forgot):
`git commit --amend`

Code Archaeology

Look at a branch's history:
`git log main`

`git log --graph main`

`git log --oneline`

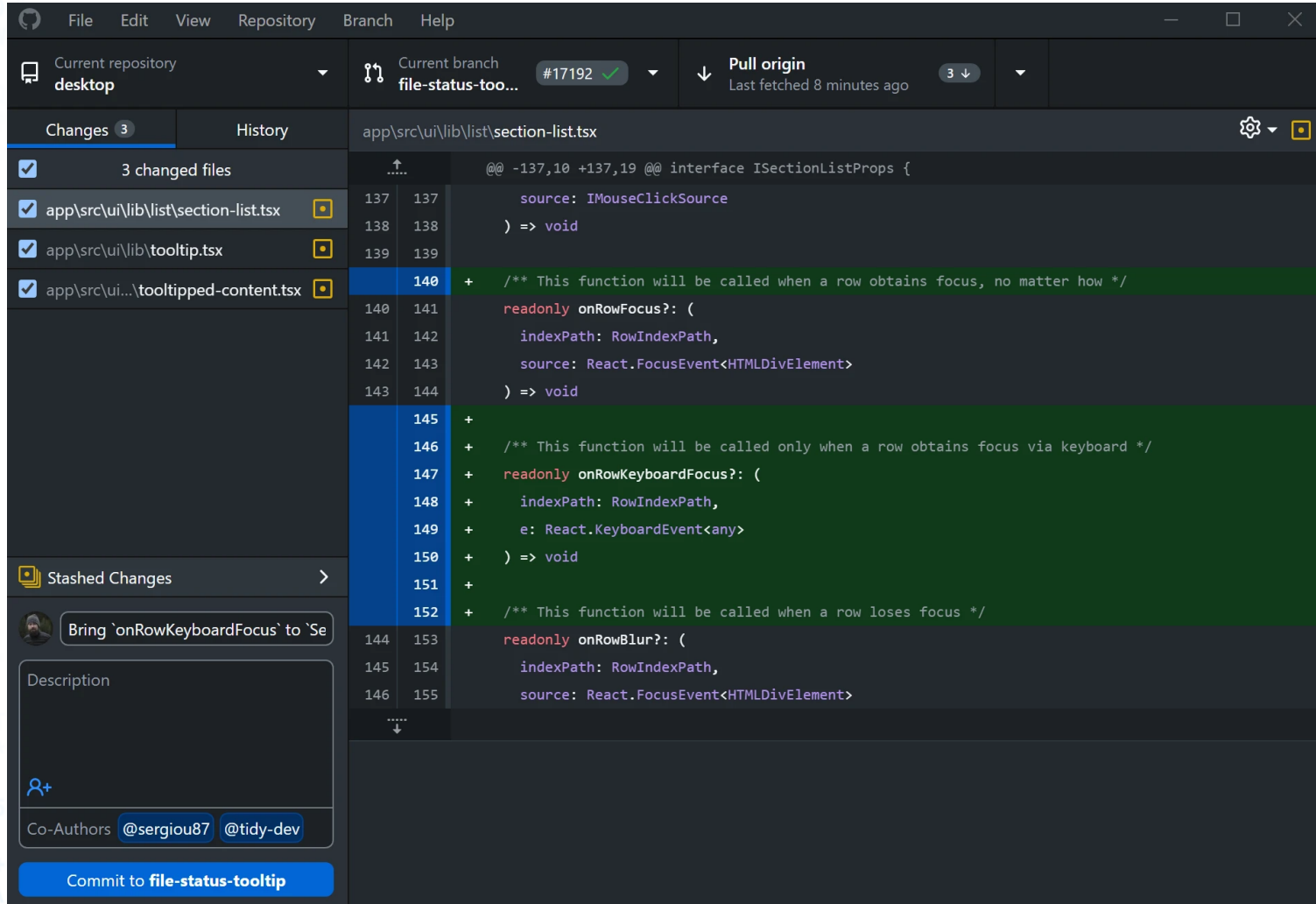
Show every commit that modified a file:
`git log <file>`

Show every commit that modified a file, including before it was renamed:
`git log --follow <file>`

Find every commit that added or removed some text:
`git log -G banana`

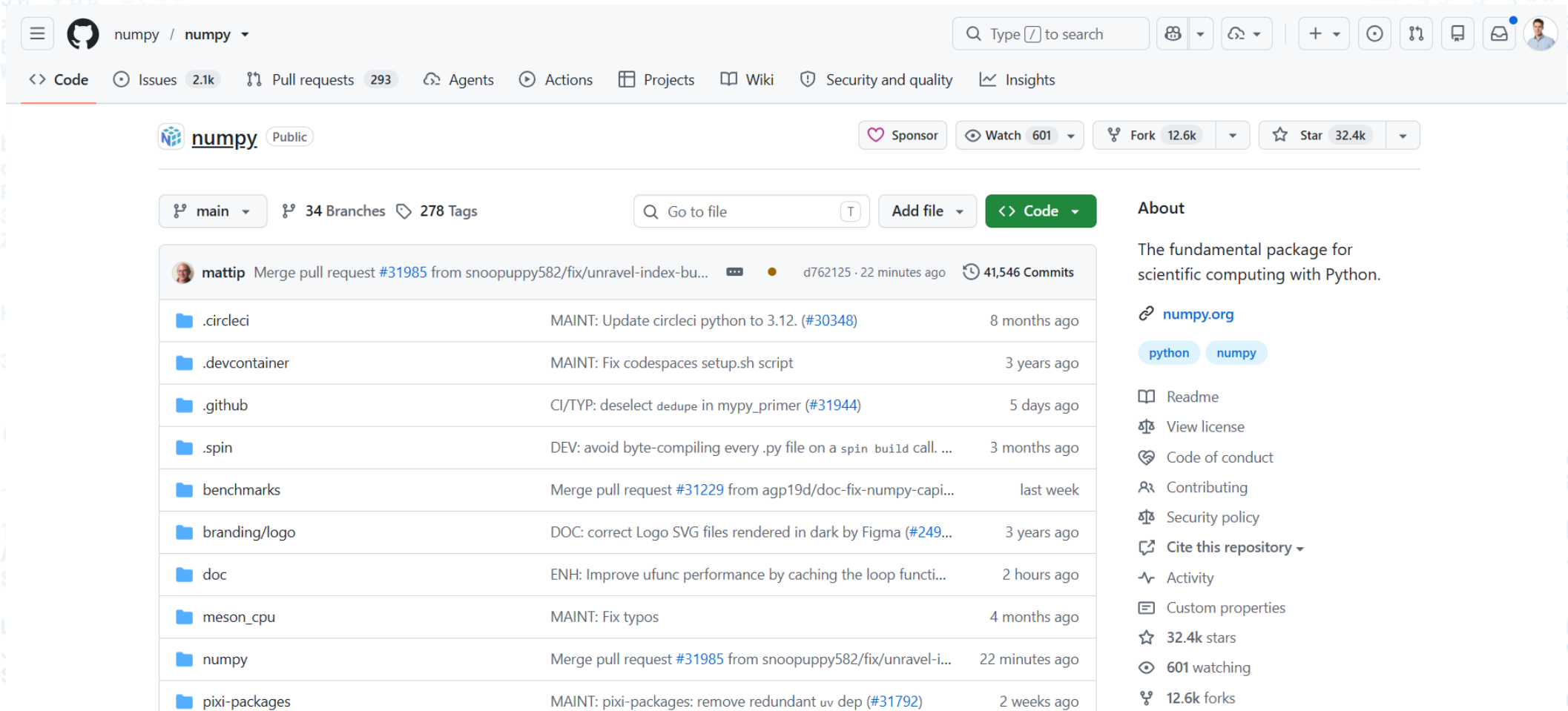
Show who last changed each line of a file:
`git blame <file>`

Version control with Git



<https://github.com/apps/desktop>, all rights reserved

Version control with Git



The screenshot shows the GitHub interface for the NumPy repository. At the top, the repository name 'numpy' is displayed with a dropdown arrow. Below this, navigation tabs include 'Code', 'Issues' (2.1k), 'Pull requests' (293), 'Agents', 'Actions', 'Projects', 'Wiki', 'Security and quality', and 'Insights'. The repository is marked as 'Public'. Action buttons for 'Sponsor', 'Watch' (601), 'Fork' (12.6k), and 'Star' (32.4k) are visible. A search bar 'Go to file' and an 'Add file' button are present. The main content area shows a list of files and folders with their commit history and timestamps. A sidebar on the right provides an 'About' section, a link to 'numpy.org', and a list of repository features like 'Readme', 'View license', 'Code of conduct', etc.

File/Folder	Commit Message	Time Ago
<code>.circleci</code>	MAINT: Update circleci python to 3.12. (#30348)	8 months ago
<code>.devcontainer</code>	MAINT: Fix codespaces setup.sh script	3 years ago
<code>.github</code>	CI/TYP: deselect dedupe in mypy_primer (#31944)	5 days ago
<code>.spin</code>	DEV: avoid byte-compiling every .py file on a spin build call. ...	3 months ago
<code>benchmarks</code>	Merge pull request #31229 from agp19d/doc-fix-numpy-capi...	last week
<code>branding/logo</code>	DOC: correct Logo SVG files rendered in dark by Figma (#249...	3 years ago
<code>doc</code>	ENH: Improve ufunc performance by caching the loop functi...	2 hours ago
<code>meson_cpu</code>	MAINT: Fix typos	4 months ago
<code>numpy</code>	Merge pull request #31985 from snoopuppy582/fix/unravel-i...	22 minutes ago
<code>pixi-packages</code>	MAINT: pixi-packages: remove redundant uv dep (#31792)	2 weeks ago

About
The fundamental package for scientific computing with Python.

numpy.org

[python](#) [numpy](#)

- Readme
- View license
- Code of conduct
- Contributing
- Security policy
- Cite this repository
- Activity
- Custom properties
- 32.4k stars
- 601 watching
- 12.6k forks

Version control with Git

